



VIT
Vellore Institute of Technology

Course Code: BECE102L

Course Name: Digital Systems Design Lab

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Class ID: 0498

Slot: L13+L14

CASE STUDY REPORT
ON
CAR PARKING SYSTEM

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1. Problem Identification

- The urban environment is facing severe problems: vehicle congestion, unauthorized parking, and inefficient manual management of car parking lots.
- There is an urgent need to look for a secure, efficient, and automated car parking system that will ease the congestion in vehicle entry and exit, improve security, and lessen operational costs.

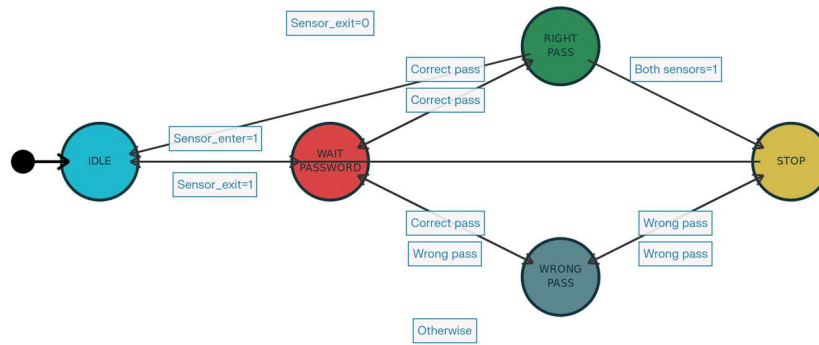


- The key challenge is to design a digital system capable of monitoring parking lot occupancy, controlling access, and providing real-time feedback for users and administrators.

2. System Design

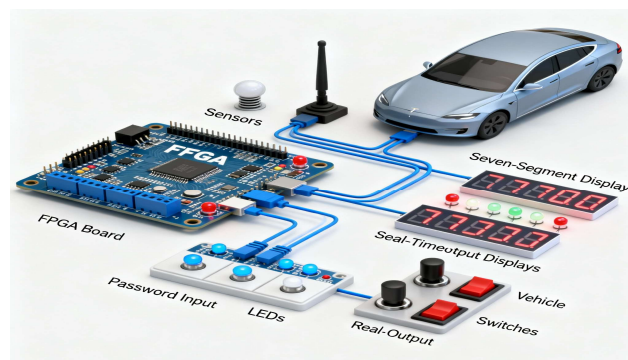
- The proposed digital car parking system is an FPGA-based design and has been implemented in Verilog HDL.
- A finite state machine controls all the operations, entrance/exit sensors detect vehicle movement, and password authentication ensures extra security in the system.
- LED and seven-segment displays are used to provide visual feedback from the system, while modular architecture helps in the scaling of the system for multiple parking zones.

Car Parking System States



3. Component Selection

- This design adopts the FPGA development board for hardware implementation, offering flexibility.
- The entrance and exit use sensors that can provide real-time detection of vehicles.
- Integrated circuits related to LED driving and seven-segment displays are chosen with compatibility with the FPGA and low cost.
- Components were justified if they could support reprogrammable logic, be cost-effective, and easy to integrate.
- Verilog HDL with the FPGA platform offers low power dissipation and fast reconfiguration, which makes this solution economically and technically strong.

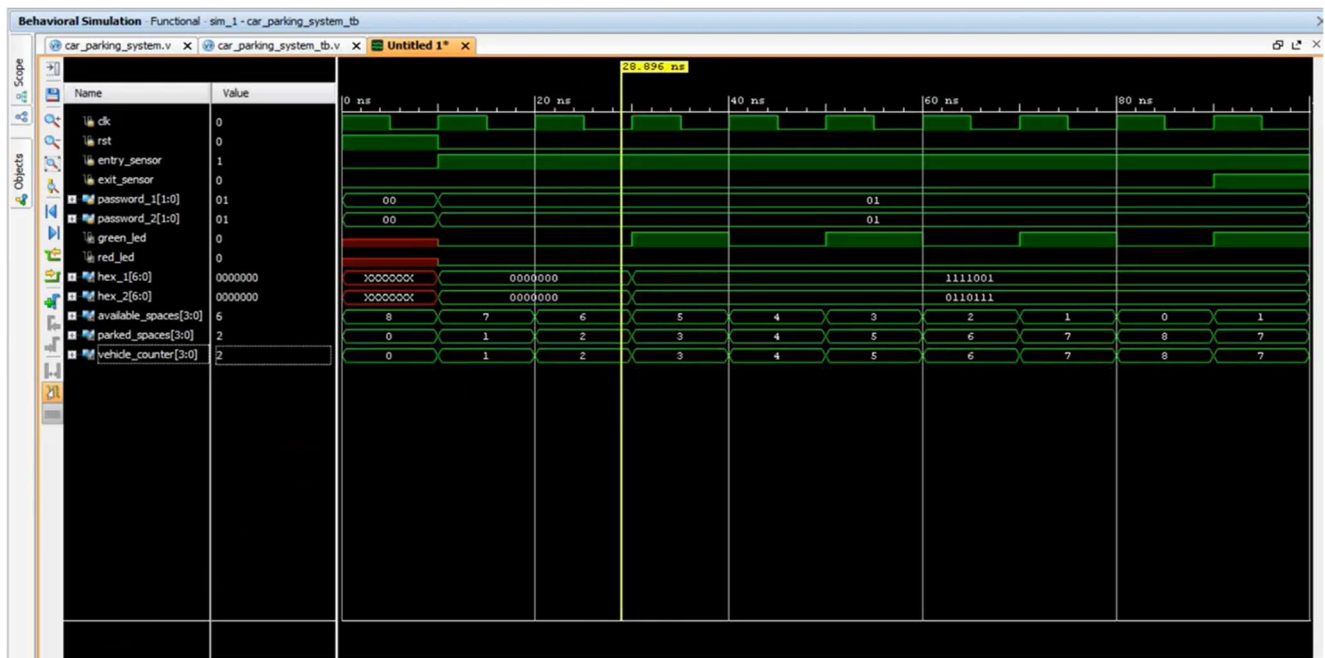


4. Implementation

- The parking system is developed with Verilog HDL, synthesized onto an FPGA platform. The architecture consists of basic vehicle detection through sensors, password verification logic, and output driving for LEDs and displays.
- A finite state machine is encoded to operate on the major states of the system, such as idle, password entry, authentication success/failure, and obstruction handling.
- Physical testing on actual switch-based inputs is demonstrated by real-time output indications on the FPGA board.

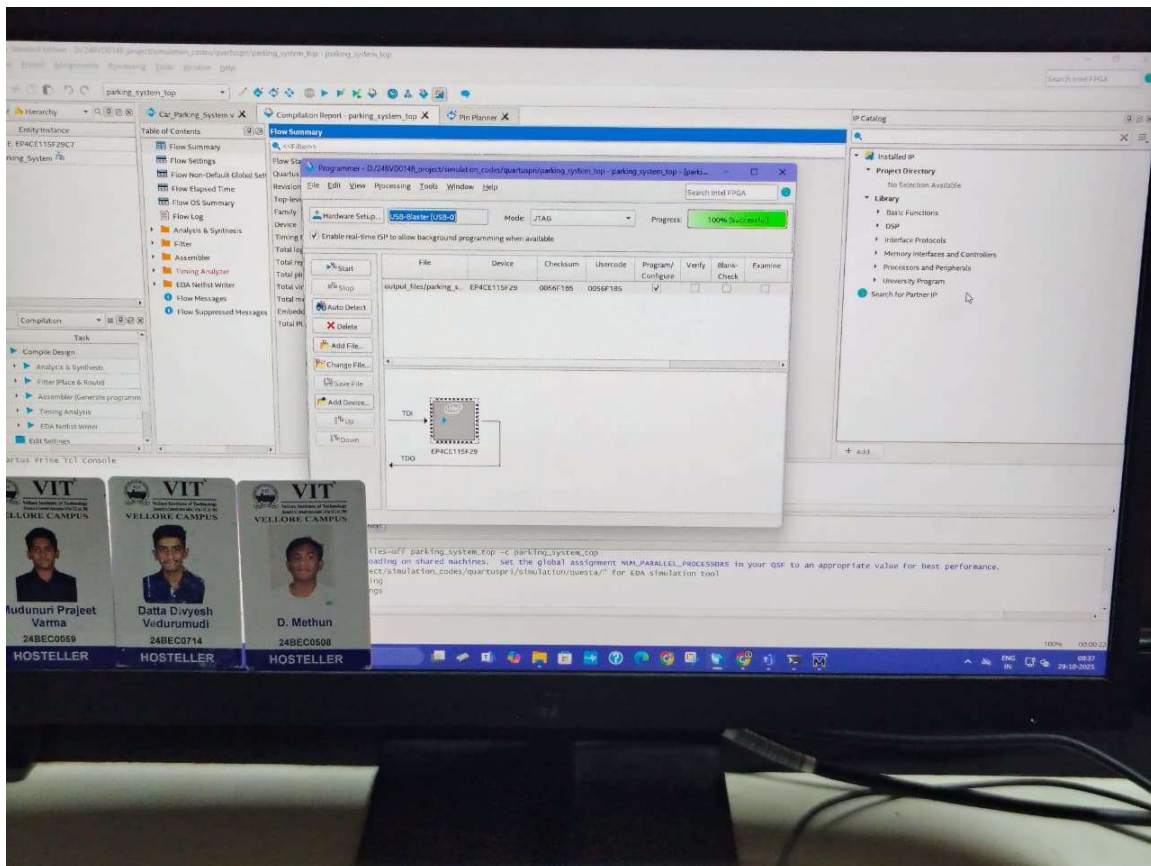
5. Testing and Verification

- The simulation of this system in ModelSim checks on state transitions, sensor input, and password logic against their respective expected outcomes.
- Actual hardware testing to confirm the proper synchronization, glitch-free transitions, correct password validation, and appropriate LED/display responses are made.
- Waveform analysis checks the behavior of the signals throughout the operation of the system, and debugging resolves any discrepancies during integration.



6. Analysis and Optimization

Efficiency is realized through synchronous design principles, keeping the use of hardware resources on the FPGA to a minimum. The module-based nature of its Verilog code makes the extension to multiple zones rather easy, with maintenance also simplified by reprogrammable logic. Reliability is assured through asynchronous reset, robust handling of inputs, and clear user feedback. Future optimizations could be password complexity increase, further power reduction, and enrichment of user interfacing.



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